2023 XXXII International Scientific Conference Electronics (ET) | 979-8-3503-0200-4/23/$31.00 ©2023 IEEE | DOI: 10.1109/ET59121.2023.10278929

An Elementary Emulator Based on Speech-To-Text and Text-to-Speech Technologies for Educational Purposes

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Abstract – **The article examines two innovative technologies Text-to-Speech (TTS) and Speech-to-Text (STT). They have been used to create a tool to facilitate the preparation of learning documentation in English, such as: lectures, laboratory exercises, course projects, tests. The article provides a brief description of TTS and STT technologies; analysis of existing developments; presentation of development trends, synthesizer technologies; a comparison of acoustic and language modeling; description of models that use STT technology; algorithm presentation and description of a simple application based on TTS and STT technologies. In summary, the results of the application testing are presented and requests for future development are indicated.**

Keywords – **Text-to-Speech, Speech-to-Text, acoustic model language model**

I. Introduction

The dynamic development of information technologies necessitates the creation of new specialties and the development of new disciplines. This requires quick and timely resource provision of the learning process. The main problem faced by teams of specialists engaged in the preparation of interactive materials for qualified training is the manual creation and processing of text documents in a short time. The possible solution is the use of Text-to-Speech (TTS) and Speech-to-Text (STT) technologies to design and develop a tool to help the teacher in the preparation of lectures, exercises, tests, etc. The advantage of such development is saving time, increasing productivity and improving the quality of the compiled documents. TTS and STT technologies used to emulate human speech have proven themselves over time and are widely used today in a number of fields such as: education, healthcare (medical documentation, therapy), telecommunications (telephony), information technology, automotive, aviation (in pilot training), financial services and insurance, e-commerce and a number of others.

II. Description of Technologies Text-to-Speech and Speech-to-Text

Text-to-Speech and speech recognition are based on two technologies: TTS and STT.

1. Text-to-Speech technologies

Text-to-Speech (TTS) [1] is a "read aloud" technology. At technology core is the conversion process, in which text is input and speech is output. The operation of the read aloud system (Text-to-Speech) is presented in Fig.1.

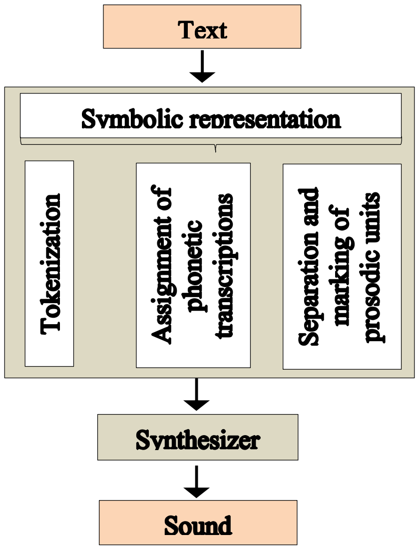

Fig. 1. Read aloud system operation

A. Description of the Read aloud system

TTS is a two-step text analysis process. The first stage defines the basic linguistic structure, including the sequence of phonemes, stress and syntactic structure important for speaking the text. The second stage converts TTS generated synthetic speech from a linguistic description. [1]

The technology is related to several concepts:

- *Tokenization* - Converting text into words.
- *Assignment of phonetic transcriptions* - Converting text to phoneme/ grapheme to phoneme. Separation and marking of prosodic units (phrase, expression and sentence).
- *Symbolic representation* - a set of tokenization and assignment of phonetic transcriptions.
- *Synthesizer transformation of the symbolic*

representation - Transforming the input text to output sound.

B. Basic synthesis techniques. Types of synthesis

Text-to-speech conversion is done through a synthesizer. Table 1. lists and describes the main types of synthesis and the main synthesis techniques.

TABLE 1. BASIC SYNTHESIS TECHNIQUES AND TYPES OF SYNTHESIS [2]

Basic synthesis techniques		
ST		Descriptions
AS	-	It applies a mechanical and acoustic model.
FS	-	No human speech samples are used at runtime; the synthesized speech output uses additive synthesis and an acoustic model.
CS	USS	Uses large speech DBs.
	DS	Entries of words and phrases that are linked to create complete uses.
	D-SS	Uses minimal speech DBs with diphones.
Types of Synthesis		
SN		Description
AS		Computational speech synthesis techniques based on models of the human vocal tract and the resulting articulation processes.
HMMBS		Based on HMM, known as statistical parametric synthesis.
SS		A technique for synthesizing speech by substitution.
ADF		DNN based deep neural networks.

Legend: ST-Synthesizer technologies, AS-Articulator synthesis, FS-Formant synthesis, CS-Concatenation synthesis, USS-Unit selection synthesis, DS-Diphonic synthesis, D-SS-Domain-Specific synthesis, SN-Synthesis name, AS-Acoustic synthesis, HMMBS-Hidden Markov models based synthesis, SS-Sinusoidal synthesis, ADF-Audio deep fakes

2. Speech-to-Text technologies

Speech-to-Text (STT) is a technology for converting "Speech-to-Text", or computer recognition or Automatic Speech Recognition (ASR) (Fig.2).

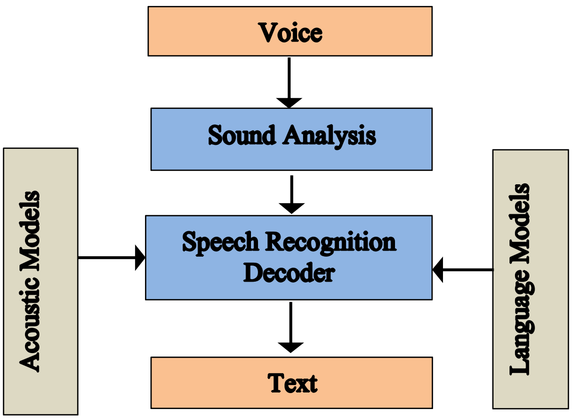

Fig. 2. Operation of the speech recognition system

A. Comparison between Acoustic Models and Language Models

TABLE 2. COMPARISON BETWEEN ACOUSTIC AND LINGUISTIC MODELING [2]

D	Model	Description
AM	The model is trained from a set of audio recordings and their corresponding transcriptions. It is created by audio recordings of speech and their textual transcriptions and using software to create statistical representations of the sounds by composing each word.	Represents the relationship between an audio signal and phonemes or other linguistic units making up speech.
LM	Given a sequence of length M, the model assigns a probability P (word$_1$, word$_2$, ...,word$_n$) to the entire sequence. Language models generate probabilities by training on text corpora in one or more languages.	Represents a probability distribution over a sequence of words.

Legend: D-Designation, AM-Acoustic modeling, LM-Language modeling

Voice recognition is done through various modern algorithms, which are based on acoustic and language modeling. A comparative analysis of these two models is presented in Table 2.

B. Commonly used models applying STT technology

- HMM - statistical Markov model, for which the modeled system is assumed to be a Markov process called X with hidden states. [3]
- Speech recognition based on Dynamic Time Warping (DTW) - this is a method to find an optimal match between two given sequences with certain enclosures. [3]
- Neural networks are an Acoustic Modeling Approach (ASR). [3]
- Deep feed-forward and feedback (recurrent) Neural Networks. [3]
- End-to-end ASR - traditional phonetic-based models require pronunciation, acoustic and language model components and training. [4]

III. ANALYSIS OF TTS AND STT TECHNOLOGY

A. Development trends

Market studies of applications using TTS and STT technologies show a trend towards a three-fold increase in STT API sales for the period from 2018 to 2030 year [5], as well as a 13.8% increase in total TTS revenues for the period from 2022 to 2029. to reach USD 6.39 billion [6].

Table 3 presents a ranking of applications using STT and TTS technologies. Among them, the most used and oldest speech recognition system is Dragon Naturally Speaking.

TABLE 3. RANKINGS OF APPLICATIONS USING STT AND TTS TECHNOLOGIES

1) Among the ten most used applications that implement STT or TTS technology [7]	
STT	TTS
AWS Amazon Transcribe AssemblyAI Deepgra NeuralSpace Resemble.AI Otter.Ai	Murf.Ai WellSaid Labs Lovo BeyondWords
2) The best STT applications for 2023 [8]	
Dragon Anywhere, Dragon Professional, Otter, Verbit, and others.	
3) Applications based on STT technology [9]	
iPhone mobile application	App for Android
Natural Reader Text-to-Speech! Speechify Text and others.	Narrator's Voice Speech Central and others.
4) The top 7 applications for synthesized speech by 2023 based on TTS technology [10]	

Google Cloud Speech-to-Text, Amazon Transcribe (Mac, Windows, Linux), AssemblyAI (Mac, Windows, Linux), IBM Watson (Linux), Rev.ai, Amberscript, Soniox, Whisper AI, One AI.
5) The best dictation software [11]
Apple Dictation - software for dictation on Apple devices (iOS, iPadOS, macOS) - free; Windows 10 Speech Recognition - Windows Dictation Software - Free; Dragon by Nuance - custom dictation app (Android, iOS, macOS, Windows); Gboard - mobile application for dictation (Android, iOS) - free; Google Docs voice typing (Web on Chrome); SpeechTexter - casual use app (Web on Chrome, Android)
6) Top 10 transcription and captioning tools [12]
Amberscript (PC, mobile version), Happy Scribe, Sonix, Amara, Ooona, Subtitle Next, Konch, Media.io, Subly
7) Top 11 best Recognition Software [13]
Dragon Professional; Dragon Anywhere; Google Now; Google Cloud Speech API; Google Docs Voice Typing; Siri; Amazon Lex; Microsoft Bing Speech API; Cortana; Voice Finger; Philips SpeechLive

B. Areas of application TTS and STT technologies

A selection and analysis of 30 publications in the IEEE Digital Library for the period 2013 - 2023 (10 years) in the field of TTS and STT technologies found that a large number of these articles comment on advances aimed at helping a particular group of society that is disadvantaged, namely people with disabilities: [14], [15], [16], [17], [18], [19], [20], [21], [22];

These technologies are the basis of developments whose purpose is related to:

- *access to various applications via voice control/ supports:*
  - to tourism semantic website [23];
  - electronic commerce application management [24];
  - voice support applications for IoT devices that enhance classroom activity [25];
  - text-to-speech Technology-Based Programming Tool [22].
- *language translation:*
  - machine translation of English videos into Indian regional languages using open innovation [26];
  - a mobile-based bilingual scene-to-speech application [27];
  - implementation of Bangla speech recognition in the Voice Input and Speech Output (viso) calculator [28];
  - user interface design for the language translator module in Swasthya slate (an m-Health tool) [29].
- *urban air mobility contingency management/cyber security:*
  - autonomous management of contingencies in urban air mobility [30];
  - virtual assistant for cyberdisease care [31].
- *speech recognition/ emotion-based communication tracking:*
  - creation of a remote computer control system using speech recognition technologies on mobile devices [32];
  - eye tracking and emotion-based speech-driven human-avatar communication [33];
- *creation of virtual assistants to help the student:*
  - a virtual assistant chatbot for a college advising system [34].
- *training:*
  - determining the reading comprehension competence of Korean students by assessing objective reading competence [35];
  - InstruMentor: An interactive robot for teaching musical instruments [36];
  - for teacher training [37].
- *Text-to-Speech conversion:*
  - speech-to-text conversion for multilingual languages, which is achieved by a deep convolutional neural network [38];
  - speech-to-text implementation using a hidden Markov model [39];
  - communications using a speech-to-Text-to-Speech pipeline [40].
- *as well as for a number of other activities such as:*
  - popularity classification of twitter messages using audio processing [41];
  - FOSD-based non-zero-origin speech data identification [42];
  - analyzing the end-to-end conversion rate of fpt [43];
  - analyzing the possible applications of B.A.R.I.C.A. to smart mobility [44].

The analysis shows that TTS and STT are innovative and widely applicable technologies. The many developments based on these technologies make it possible to increase revenue and increase business efficiency.

IV. Design and Development of an Application for Creating Laboratory Exercises, Lectures and English Language Tests Using TTS and STT Technologies

A. An elementary emulator based on STT and TTS technologies for educational purpose (EEbSttTtsT) - algorithmic description

1). Start the program
2). Select of TTS
2.1. From Setting File, select a text file (*.txt, *.docx or *.pdf)
2.2. From Settings Voice, select a voice to read the text - Male or Female, by adjusting the Speed and Volume of the sound.
2.3. Turn on automatic text reading (Read Text button)
2.4. Stop automatic reading tex (Pause button)
2.5. Continue automatic reading text (Resume button)
2.6. Stop automatic reading text (Stop button)
2.7. Saving text (button Save As File *.wav)
*Note: The text can be saved both in *.wav and in another format, such as (*.txt, *.docx)*

3). Selection – STT
3.1. Start Speech-to-Text (Speech text button)
3.2. Termination of STT recording (Stop text button)
3.3. Saving text (button Save As File *.txt, *.docx)
Note: Using the Speaker recording check button, options are visualized: Record, Save Record and Execute buttons:
3.1. Start recording speech (Record button)
3.2. Saving speech (Save Record button, recording in *.wav format)
3.3. Listening to the speech recording (Execute button)

B. Diagrams, implementation, libraries, classes and methods

The diagrams in Fig.3 and Fig.4 describe the options available to the user when selecting TTS and STT.

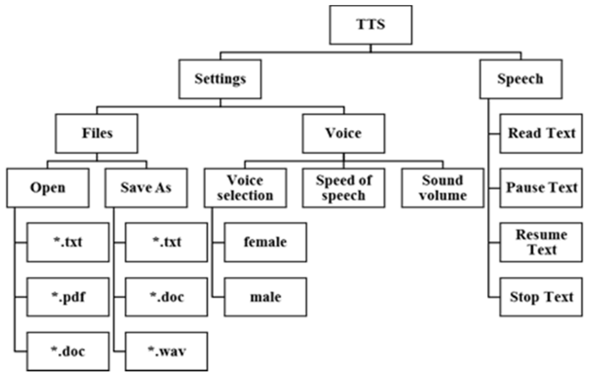

Fig. 3. Diagram - selection TTS

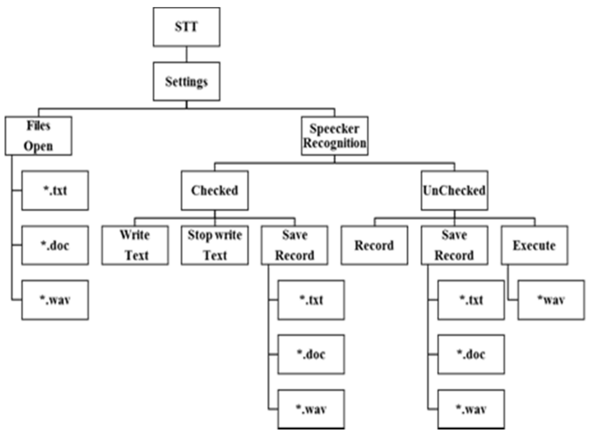

Fig. 4. Diagram - selection STT

C# programming language and Visual Studio 2022 Community development environment were used for the implementation of the elementary application. The following libraries and methods were used to develop a rudimentary emulator based on speech-to-text and text-to-speech technologies and are reflected in Table 4:

TABLE 4. LIBRARIES AND METHODS USED IN THE DEVELOPMENT OF AN EEbSttTtsT

Libraries	Class
- Microsoft.Office.Interop.Word - for working with *.docx documents; - IO - for working with text files; - org.apache.pdfbox.pdmodel - for working with pdf; - org.apache.pdfbox.util - for working with pdf; - Speech – provides access and control of the speech recognition engine in Prozess; - Globalization - required to create a SpeechRecognitionEngine object. - winmm.dll;	- SpeechSynthesizer; - PromptBuilder; - SpeechRecognitionEngine; - saveFileDialog; - OpenFileDialog; - StreamReader; - PDDocument; - PDFTextStripper; - Grammar; - Choices;
Methods for TTS	Methods for STT
- void ChoiceVoice - for voice selection (male/female); - void Pause – stops reading aloud; - void Resume – continues reading aloud; - void Stop – stops reading aloud; - void SaveAsFile – saves a text file in audio format - *.wav; - void OpenFile – opens a text file for reading; - Speech.Synthesis – to initialize and configure the speech (or voice) synthesis engine that converts text to audio (or Text-to-Speech);	- void WriteText – write Speech-to-Text; - void StopWriteText – stops writing Speech-to-Text; - void SaveRecordFile - saves speech in a text file -*.txt, *.docx; - void read() – loads the resource local text file; Methods for STT *(the additional options)* - void Record – voice recording; - void SaveRecordWav – saves voice in audio format - *.wav; - void Execute – executes an audio recording in *.wav format. - Speech.Recognition - to identify spoken words and convert them into readable text (speech or Speech-to-Text);

C. Screen forms of Application for creating laboratory exercises and lectures in English based on TTS and STT technology

After starting the application, there are two options (Fig.5): TTS and STT.

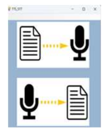

Fig. 5. Home screen form

When TTS is selected, the screen form shown in Fig. 6 is visualized.

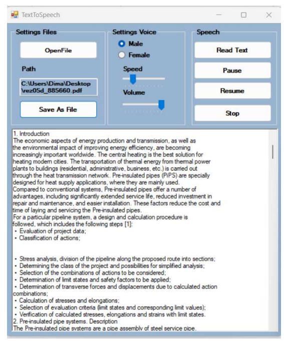

Fig. 6. Screenshot of TTS

When selecting STT, the screens form shown in Fig.7 and Fig. 8 is visualized.

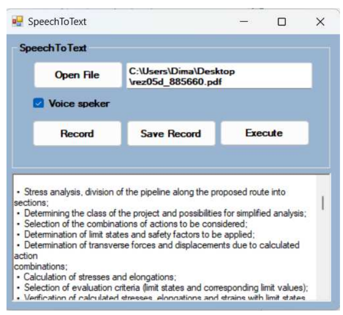

Fig. 7. Screenshot of STT with speaker unchecked recognition option

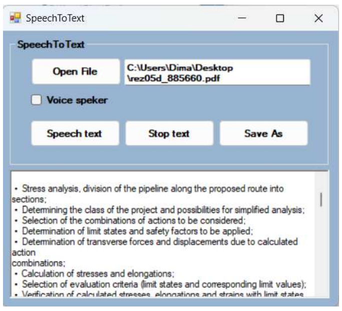

Fig. 8. Screenshot of STT with speaker checked recognition option

D. The test results of the application are summarized in Table 5. They were made on the basis of 3 groups of criteria:

- Access to resources;
- Sound characteristics;
- Speech recording speed and pronunciation.

TABLE 5. TTS AND STT BASED APPLICATION TESTS

Test	Resource	TTS	STT
Access to resources	Internet access	It is not necessary	It is not necessary
	Local word dictionary	It is not necessary	Yes, it is necessary
Sound characteristics	Sound volume	No problems found	No problems found
	Speed of sound	No problems found	No problems found
Speech recording	Word write speed	No problems found	Recording delay
	Pronunciation of a word	No problems found	Incorrect spelling with incorrect pronunciation

The conclusions of the tests made regarding the application are:

1. In terms of resources, Internet is not required for both technologies (STT and TTS), a local dictionary is not required for TTS, unlike STT where one is required (i.e. English word file).
2. Regarding the characteristics of sound speed, no problems were found.
3. In terms of Text-to-Speech recording, TTS recording speed and pronunciation are not affected by this technology, while STT has a delay in recording words. Incorrect spelling is also noted when pronunciation is incorrect.

V. CONCLUSION

Today, many businesses and individuals use STT and TTS APIs for various purposes such as: dictation, chat bots, translations, transcription and a number of others. The main objective of the paper was to explore TTS and STT technologies and to design and develop an application to help teachers guide English language teaching and in particular to create lectures, labs and more in English. The paper described a read-aloud system, presented the requirements for a synthetic speech system, and outlined the main synthesizer techniques and their characteristics, as well as the main types of synthesis. The article discussed STT, comparing the acoustic and linguistic models underlying this technology. The presented development and growth trends according to different rankings proved that TTS and STT is a widely applied and researched field. The presented classification of application areas made for the period 2013-2023 based on TTS or STT technologies, confirms the usefulness of these technologies in modern life. The elementary emulator based on STT and TTS technologies for educational purposes will help teachers by facilitating their activities and shortening the development time of lectures, demonstrations, tests and more. The application was tested on 3 metrics, which are an indicator of independence in terms of the basic resources required for proper functioning.

In conclusion, work on the application continues by adding additional options and multilingualism to the application and improving the existing ones.

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